

TAREA:

Galera 4 Clúster con MariaDB en Linux

ASIGNATURA:

Cómputo de alto desempeño

ALUMNA:

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PROGRAMA EDUCATIVO:

Ingeniería en Datos e Inteligencia Organizacional

Introducción

En este reporte se detalla el proceso de creación de un clúster en Ubuntu Server utilizando Galera 4 y la realización de pruebas de rendimiento con Sysbench. El clúster de base de datos está implementado con MariaDB.

Software

- Virtual box
- Imagen ISO de ubuntu server
- MariaDB
- Mysql
- Galera 4
- Sysbench
- Open-ssh
- CMD de Windows: En este fueron ejecutados los comandos.

Pasos Realizados

1. Instalación del Servidor Ubuntu Server:

- Asignación de Recursos: Se asignó un core para el servidor Ubuntu Server.
- Configuración de la Red:
 - Red NAT: Se configuró la red NAT para permitir la descarga de paquetes necesarios.
 - Port Forwarding: Se configuró el port forwarding para habilitar la conexión a través de SSH.

2. Configuración Inicial del Sistema:

- Instalación de OpenSSH: Se habilitó y descargó el módulo OpenSSH para permitir la conexión externa a la máquina virtual:
 - `sudo apt install openssh-server`
- Actualización de Repositorios: Se actualizó la lista de repositorios del servidor:
 - `sudo apt update`
- Instalación de Paquetes Necesarios: Se instalaron los siguientes paquetes:
 - `sudo apt -y install net-tools`

```
Oracle VM VirtualBox Administrator
root@UbuntuServer: /home/v \ x + v
Reading state information... Done
151 packages can be upgraded. Run 'apt list --upgradable' to see them.
root@UbuntuServer: /home/vboxuser# apt install net-tools
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  net-tools
0 upgraded, 1 newly installed, 0 to remove and 151 not upgraded.
Need to get 204 kB of archives.
After this operation, 811 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble/main amd64 net-tools amd64 2.10-0.1ubuntu4 [204
kB]
Fetched 204 kB in 1s (281 kB/s)
Selecting previously unselected package net-tools.
(Reading database ... 83845 files and directories currently installed.)
Preparing to unpack .../net-tools.2.10-0.1ubuntu4_amd64.deb ...
Unpacking net-tools (2.10-0.1ubuntu4) ...
Setting up net-tools (2.10-0.1ubuntu4) ...
Processing triggers for man-db (2.12.0-4build2) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
root@UbuntuServer: /home/vboxuser# apt -y install net-tools
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
net-tools is already the newest version (2.10-0.1ubuntu4).
0 upgraded, 0 newly installed, 0 to remove and 151 not upgraded.
root@UbuntuServer: /home/vboxuser#
```

- sudo apt -y install software-properties-common

```
Oracle VM VirtualBox Administrator
root@UbuntuServer: /home/v \ x + v
Do you want to continue? [Y/n] ^C
root@UbuntuServer: /home/vboxuser# apt -y install software-properties-common
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  python3-software-properties
The following packages will be upgraded:
  python3-software-properties software-properties-common
2 upgraded, 0 newly installed, 0 to remove and 149 not upgraded.
Need to get 44.1 kB of archives.
After this operation, 0 B of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 software-properties-common a
ll 0.99.49.1 [14.4 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 python3-software-properties
all 0.99.49.1 [29.7 kB]
Fetched 44.1 kB in 1s (60.7 kB/s)
(Reading database ... 83893 files and directories currently installed.)
Preparing to unpack .../software-properties-common_0.99.49.1_all.deb ...
Unpacking software-properties-common (0.99.49.1) over (0.99.48) ...
Preparing to unpack .../python3-software-properties_0.99.49.1_all.deb ...
Unpacking python3-software-properties (0.99.49.1) over (0.99.48) ...
Setting up python3-software-properties (0.99.49.1) ...
Setting up software-properties-common (0.99.49.1) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for dbus (1.14.10-4ubuntu4.1) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
root@UbuntuServer: /home/vboxuser#
```

3. Instalación y Configuración de Galera 4:

- Instalación de MariaDB y Galera 4: Se instalaron los paquetes necesarios:
 - sudo apt -y install mariadb-server mariadb-client galera-4

```
root@UbuntuServer: /home/v# apt list --upgradable to see them.
root@UbuntuServer: /home/vboxuser# apt -y install mariadb-server mariadb-client galera-4
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libcgi-fast-perl libcgi-pm-perl libclone-perl libconfig-inifiles-perl libdbd-mysql-perl
  libdbi-perl libencode-locale-perl libfcgi-bin libfcgi-perl libfcgi0t64
  libhtml-parser-perl libhtml-tagset-perl libhtml-template-perl libhttp-date-perl
  libhttp-message-perl libio-html-perl liblwp-mediatypes-perl libmariadb3 libmysqlclient21
  libsnappy1v5 libtime-date-perl liburi-perl liburing2 mariadb-client-core mariadb-common
  mariadb-plugin-provider-bzip2 mariadb-plugin-provider-lz4 mariadb-plugin-provider-lzma
  mariadb-plugin-provider-lzo mariadb-plugin-provider-snappy mariadb-server-core
  mysql-common pv socat
Suggested packages:
  libdbd-m-perl libnet-daemon-perl libsql-statement-perl libdata-dump-perl
  libipc-sharedcache-perl libio-compress-brotli-perl libbusiness-isbn-perl
  libregexp-ipv6-perl libwww-perl mailx mariadb-test doc-base
The following NEW packages will be installed:
  galera-4 libcgi-fast-perl libcgi-pm-perl libclone-perl libconfig-inifiles-perl
  libdbd-mysql-perl libdbi-perl libencode-locale-perl libfcgi-bin libfcgi-perl libfcgi0t64
  libhtml-parser-perl libhtml-tagset-perl libhtml-template-perl libhttp-date-perl
  libhttp-message-perl libio-html-perl liblwp-mediatypes-perl libmariadb3 libmysqlclient21
  libsnappy1v5 libtime-date-perl liburi-perl liburing2 mariadb-client-core mariadb-client-core
  mariadb-common mariadb-plugin-provider-bzip2 mariadb-plugin-provider-lz4
  mariadb-plugin-provider-lzma mariadb-plugin-provider-lzo mariadb-plugin-provider-snappy
  mariadb-server mariadb-server-core mysql-common pv socat
0 upgraded, 37 newly installed, 0 to remove and 149 not upgraded.
Need to get 19.9 MB of archives.
After this operation, 197 MB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble/universe amd64 galera-4 26.4.16-2build4 [
736 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble/main amd64 mysql-common all 5.8+1.1.0build1 [6,
746 B]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 mariadb-common all 1:10.
11.8-0ubuntu0.24.04.1 [27.1 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble/main amd64 libdbi-perl amd64 1.643-4build3 [721
kB]
Get:5 http://archive.ubuntu.com/ubuntu noble/main amd64 libconfig-inifiles-perl all 3.000003
```

- sudo apt -y install galera-arbitrator-4

```
root@UbuntuServer: /home/v# apt -y install galera-arbitrator-4
No VM guests are running outdated hypervisor (qemu) binaries on this host.
root@UbuntuServer: /home/vboxuser# apt -y install galera-arbitrator-4
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libboost-program-options1.83.0
The following NEW packages will be installed:
  galera-arbitrator-4 libboost-program-options1.83.0
0 upgraded, 2 newly installed, 0 to remove and 149 not upgraded.
Need to get 896 kB of archives.
After this operation, 4,046 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libboost-program-options1.83
.0 amd64 1.83.0-2.1ubuntu3.1 [320 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble/universe amd64 galera-arbitrator-4 amd64 26.4.1
6-2build4 [575 kB]
Fetched 896 kB in 1s (689 kB/s)
Selecting previously unselected package libboost-program-options1.83.0:amd64.
(Reading database ... 84846 files and directories currently installed.)
Preparing to unpack .../libboost-program-options1.83.0-2.1ubuntu3.1_amd64.deb ...
Unpacking libboost-program-options1.83.0:amd64 (1.83.0-2.1ubuntu3.1) ...
Selecting previously unselected package galera-arbitrator-4.
Preparing to unpack .../galera-arbitrator-4_26.4.16-2build4_amd64.deb ...
Unpacking galera-arbitrator-4 (26.4.16-2build4) ...
Setting up libboost-program-options1.83.0:amd64 (1.83.0-2.1ubuntu3.1) ...
Setting up galera-arbitrator-4 (26.4.16-2build4) ...
Created symlink /etc/systemd/system/garbd.service → /usr/lib/systemd/system/garb.service.
Created symlink /etc/systemd/system/multi-user.target.wants/garb.service → /usr/lib/systemd/
system/garb.service.
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.3) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.
```

- sudo apt -y install mariadb-client libmariadb3

```
root@UbuntuServer: /home/vboxuser# apt -y install mariadb-client libmariadb3
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
mariadb-client is already the newest version (1:10.11.8-0ubuntu0.24.04.1).
libmariadb3 is already the newest version (1:10.11.8-0ubuntu0.24.04.1).
libmariadb3 set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 149 not upgraded.
root@UbuntuServer: /home/vboxuser#
```

4. Configuración del Clúster:

- Archivo de Configuración: Nos movemos a la siguiente ruta y reemplazamos el contenido del archivo:

- `vi /etc/mysql/mariadb.conf.d/60-galera.cnf`

```
[mysqld]
binlog_format=ROW
default-storage-engine=innodb
innodb_autoinc_lock_mode=2
bind-address=0.0.0.0

# Galera Provider Configuration
wsrep_on=ON
wsrep_provider=/usr/lib/galera/libgalera_smm.so

# Galera Cluster Configuration
wsrep_cluster_name="test_cluster"
wsrep_cluster_address="gcomm://192.168.56.101"

# Galera Synchronization Configuration
wsrep_sst_method=rsync

# Galera Node Configuration
wsrep_node_address="192.168.56.101"
wsrep_node_name="nodo1"
```

- Iniciar el Clúster: Se ejecuta el siguiente comando para iniciar el clúster:

- `galera_new_cluster`

```
root@UbuntuServer:/home/vboxuser# galera_new_cluster
root@UbuntuServer:/home/vboxuser# netstat -tlnp
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State       PID/Program
tcp        0      0 0.0.0.0:4567             0.0.0.0:*               LISTEN      4351/mariadb
tcp        0      0 127.0.0.53:53           0.0.0.0:*               LISTEN      424/systemd-
tcp        0      0 127.0.0.54:53           0.0.0.0:*               LISTEN      424/systemd-
tcp        0      0 0.0.0.0:3306            0.0.0.0:*               LISTEN      4351/mariadb
tcp6       0      0 :::22                   :::*                    LISTEN      1/init
root@UbuntuServer:/home/vboxuser# mysql -u root -p -e "SHOW STATUS LIKE 'wsrep_cluster_size'"
Enter password:
+-----+-----+
| Variable_name | Value |
+-----+-----+
| wsrep_cluster_size | 1 |
+-----+-----+
```

- Validación del Clúster: Para validar la correcta creación del clúster, ejecuté los siguientes comandos:
 - `mysql -u root -p -e "SHOW STATUS LIKE 'wsrep_cluster_size'"`
 - `mysql -u root --execute="SHOW GLOBAL STATUS WHERE Variable_name IN ('wsrep_ready', 'wsrep_cluster_size', 'wsrep_cluster_status', 'wsrep_connected');"`

```

root@UbuntuServer:/home/vboxuser# mysql -u root -p -e "SHOW STATUS LIKE 'wsrep_cluster_size'"
Enter password:
+-----+-----+
| Variable_name | Value |
+-----+-----+
| wsrep_cluster_size | 1 |
+-----+-----+
root@UbuntuServer:/home/vboxuser# mysql -u root --execute="SHOW GLOBAL STATUS WHERE Variable_name IN ('wsrep_ready', 'wsrep_cluster_size', 'wsrep_cluster_status', 'wsrep_connected');"
+-----+-----+
| Variable_name | Value |
+-----+-----+
| wsrep_cluster_size | 1 |
| wsrep_cluster_status | Primary |
| wsrep_connected | ON |
| wsrep_ready | ON |
+-----+-----+

```

- Verificación de Puertos: Se ejecutó el comando para validar los puertos correspondientes al clúster:
 - netstat -tln

```

Last login: Sun Feb 16 01:04:01 2025 from 192.168.56.1
vboxuser@UbuntuServer:~$ sudo su
[sudo] password for vboxuser:
root@UbuntuServer:/home/vboxuser# netstat -tln
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State       PID/Program
tcp        0      0 127.0.0.54:53          0.0.0.0:*                LISTEN      421/systemd-
tcp        0      0 127.0.0.53:53          0.0.0.0:*                LISTEN      421/systemd-
tcp6       0      0 :::22                  :::*                    LISTEN      1/init

```

5. Instalación de Sysbench:

- Instalación de Sysbench: Instalamos Sysbench con el siguiente comando:
 - sudo apt -y install sysbench
- Creación de la Base de Datos para Pruebas:
 - mysql -u root -p -e "create database sbtest"

```

root@UbuntuServer:/home/vboxuser# mysql -u root -p -e "create database sbtest"
Enter password:
root@UbuntuServer:/home/vboxuser# sysbench --threads=1 --db-driver=mysql --mysql-user=root --events=0 oltp_apread_only prepare
sysbench 1.0.20 (using system LuaJIT 2.1.0-beta3)

FATAL: Cannot find benchmark 'oltp_apread_only': no such built-in test, file or module

```

6. Preparación del Entorno para Pruebas:

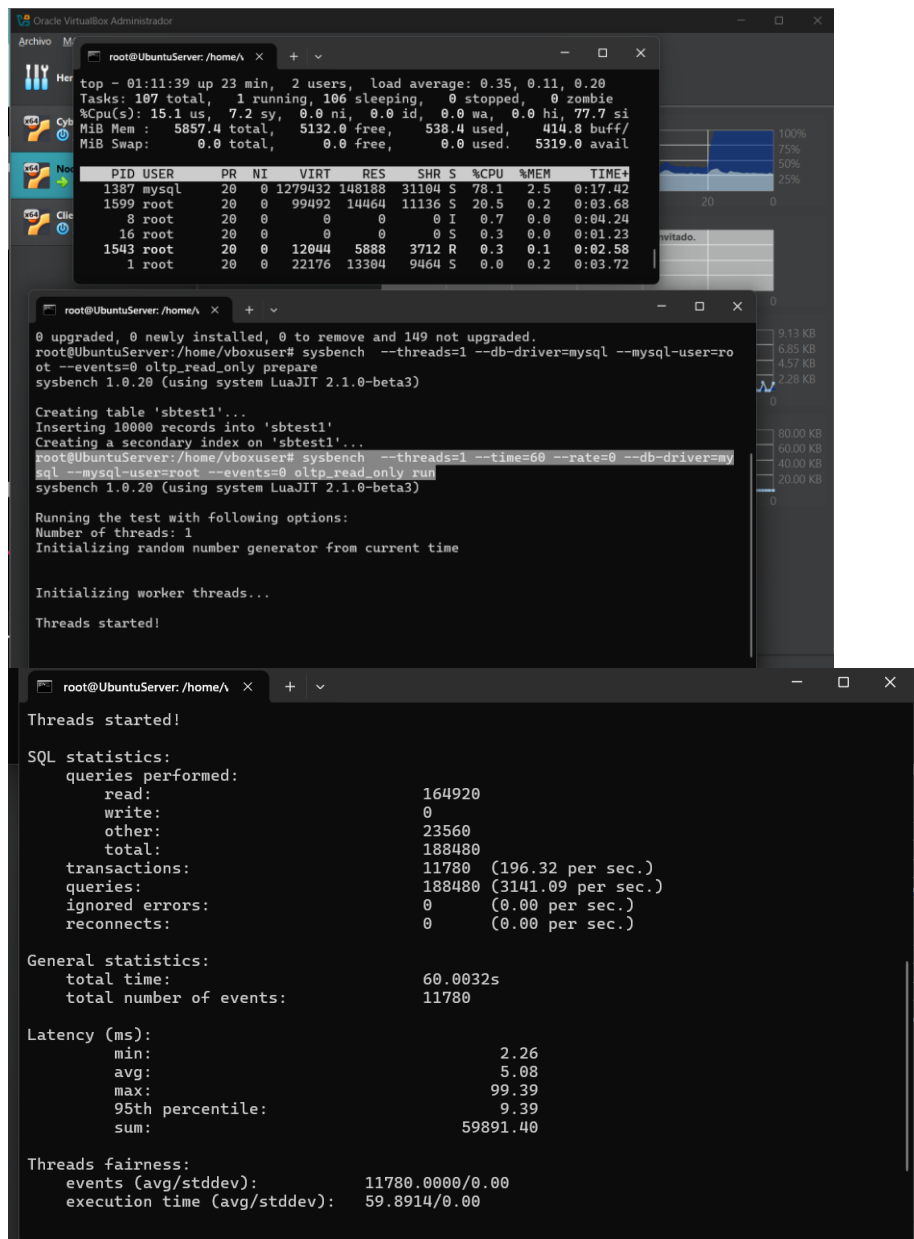
- Para realizar pruebas de solo lectura a la base de datos, ejecutamos el siguiente comando:

- sysbench --threads=1 --db-driver=mysql --mysql-user=root --events=0 oltp_read_only prepare

```
0 upgraded, 0 newly installed, 0 to remove and 149 not upgraded.
root@UbuntuServer:/home/vboxuser# sysbench --threads=1 --db-driver=mysql --mysql-user=root --events=0 oltp_read_only prepare
sysbench 1.0.20 (using system LuaJIT 2.1.0-beta3)

Creating table 'sbtest1'...
Inserting 10000 records into 'sbtest1'
Creating a secondary index on 'sbtest1'...
root@UbuntuServer:/home/vboxuser#
```

- Para ejecutar la prueba, se utiliza el siguiente comando:
 - sysbench --threads=1 --time=5 --rate=0 --db-driver=mysql --mysql-user=root --events=0 oltp_read_only run



1 CORE

Comando	Queri es totale s	Transaccio nes totales	Transaccio nes por segundo	Tiem po	Núme ro total de event os	Latencia Máxima(ms)
Oltp_read_only	1848 64	11554	192.56 por seg.	1 min	1155 4	77.09 ms
Oltp_delete	1660 64	166064	2767.64	1 min	1660 64	60.94
Oltp_insert	1085 9	10859	180.97	1 min	1085 9	47.54
oltp_point_select	3133 71	313371	5222.11	1 min	3133 71	15.54
oltp_read_write	1397 00	6985	116.40	1 min	6985	122.27
oltp_update_inde x	1232 7	12327	205.36	1 min	1232 7	499.45
oltp_update_non_ index	1319 6	13196	219.91	1 min	1319 6	514.03
oltp_write_only	4888 8	8148	135.78	1 min	8148	304.07
select_random_p oints	1384 7	13847	230.76	1 min	1384 7	455.00
select_random_ra nges	1782 3	17823	297.04	1 min	1782 3	14.62

2 CORE

Comando	Queri es totale s	Transaccio nes totales	Transaccio nes por segundo	Tiem po	Núme ro total de event os	Latencia Máxima(ms)
oltp_read_only	5513 12	34457	574.22	1 min	3445 7	221.30
oltp_delete	3676 20	367620	6124.45	1 min	3676 20	94.14
oltp_insert	6628	6628	110.43	1 min	6628	159.91
oltp_point_select	7057 06	705706	11758.79	1 min	7057 06	147.50
oltp_read_write	7997 8	3998	66.59	1 min	3998	208.75
oltp_update_inde x	8149	8149	135.77	1 min	8149	103.82
oltp_update_non_ index	7875	7875	131.19	1 min	7875	103.16
oltp_write_only	5046 6	8411	140.16	1 min	8411	138.35
select_random_p oints	7915 58	791558	13191.94	1 min	7915 58	33.57
select_random_ra nges	7855 73	785573	13091.75	1 min	7855 73	33.05

Conclusión

En el set de pruebas de 2 núcleos muestra un rendimiento significativamente mejor en términos de transacciones por segundo y latencia máxima en comparación con la de 1 núcleo, así que, el uso de más núcleos puede mejorar significativamente el rendimiento de la base de datos en escenarios de alta carga.